Biology Extended Essay: student guide

A step-by-step workbook from November of Year 1 to the viva voce in Year 2. The online version is at whatthecell.org/extended-essay.html

> **How to use this guide.** Work through it in order. Each stage tells you what to do, what to bring to your supervisor, and what to write in your researcher's reflection space (RRS). Tick each box as you finish it, and write your school's deadlines in the spaces provided.

Your timeline at a glance

STAGE	WHEN	MY DEADLINE
1. Choose a subject and supervisor	Year 1, November	
2. Explore a topic	Year 1, November to December	
3. Research question and **first reflection session**	Year 1, December	
4. Plan, safety, ethics and literature review	Year 1, January	
5. Pilot and data collection	Year 1, February to March	
6. Analysis	Year 1, March to April	
7. **Interim reflection session** and full draft	Year 1, April to early May	
8. Feedback and final essay	Year 1, May to end of June (before the summer holidays)	
9. **Viva voce** and reflective statement	Year 2, November	

The three formal reflection sessions (in bold) are recorded on your reflection and progress form (RPF). You can also ask your supervisor for short check-ins at any time. It is your job to arrange meetings.

Before you start: set up your RRS

Your researcher's reflection space is a notebook or a digital document where you record your journey. Make one entry every time you work on the EE. Each entry needs:

- the date and how long you worked
- what you did (reading, planning, experiments, writing)
- one decision you made and why
- one problem or question you still have

- sources you used, with full details

Your RRS is the raw material for your reflective statement, and it proves the work is yours.

Stage 1: Choose a subject and supervisor (November)

- ☐ Decide on the subject-focused pathway (Biology) or the interdisciplinary pathway (Biology plus one other DP subject).
- ☐ Read the “Extended Essay in Biology” page on What the Cell.
- ☐ Read at least two strong Biology EEs your supervisor shares with you. Note what makes them strong.
- ☐ Set up your RRS and your single working document for the essay.
- ☐ Meet your supervisor and agree how you will contact each other.

RRS prompt: Why Biology? What areas of biology have you enjoyed most, and why?

Stage 2: Explore a topic (November to December)

- [ ] List three topics that interest you. For each, write two sentences on why.
- [ ] Read around each topic for an hour: textbook, reviews, news articles, then journal articles.
- [ ] For each topic, note: what is known, what scientists still argue about, what you could measure.
- [ ] Check what equipment, organisms or databases the school has for each.
- [ ] Choose one topic.

RRS prompt: Which topic did you choose, and what made you drop the others?

Stage 3: Research question and first reflection session (December)

Build your question with this pattern, then test it against the checks below.

> What is the effect of **[independent variable, with its range]** on **[dependent variable, and how it is measured]** in **[organism, *Genus species*]**?
>
> or: To what extent is **[variable A]** correlated with **[variable B]** in **[population, place, time]**?

My research question:

> .....

- ☐ It asks one thing (no “and” joining two questions).
- ☐ It names the organism, variables and what is measured.
- ☐ The answer is not obvious before I start.
- ☐ It leads to analysis, not a description.
- ☐ I can answer it with school equipment, or with reliable published data, in the time I have.
- ☐ I have booked and attended the **first reflection session** with my supervisor.

Bring to the session: your topic notes, two or three draft questions, and how you plan to collect data.

RRS prompt (after the session): How did your question change during the conversation, and why?

Stage 4: Plan, safety, ethics and literature review (January)

Variables table

VARIABLE	WHAT IT IS	HOW I WILL CHANGE, MEASURE OR CONTROL IT
Independent		
Dependent		
Controlled 1		
Controlled 2		
Controlled 3		

Risk and ethics check

HAZARD OR ETHICAL ISSUE	RISK	HOW I WILL REDUCE IT

- ☐ My method uses at least five values of the independent variable, with repeats (or a large enough sample for fieldwork or database work).
- ☐ No animal will be harmed or distressed; any human participants give informed consent and stay anonymous.
- ☐ My supervisor has approved my plan, risk assessment and ethics check.
- ☐ I have found at least five reliable sources on the biology and two on the method, and recorded them fully.

Where to search: Google Scholar, PubMed, the Directory of Open Access Journals (DOAJ), your school library databases, and data sources such as GBIF or NCBI.

RRS prompt: What did the literature tell you that changed your plan?

Stage 5: Pilot and data collection (February to March)

- [ ] Run a small pilot. Note what went wrong and what you will change.
- [ ] Update the method in your working document.
- [ ] Collect all your raw data. Record it straight into a table with units and uncertainties.
- [ ] Take dated photos of your set-up and results.
- [ ] Record anything unexpected: these become your evaluation.

RRS prompt: What surprised you during data collection?

Stage 6: Analysis (March to April)

- [ ] Process the data (means, standard deviations, rates or percentages).
- [ ] Draw graphs that suit the data, with error bars where you can.
- [ ] Choose and run a statistical test that suits your data (the IA toolkit's test chooser can help).
- [ ] Write one or two sentences under every table and graph saying what it shows.

MY DATA	GRAPH	STATISTICAL TEST

RRS prompt: Do your results support what you expected? If not, why might that be?

Stage 7: Interim reflection session and full draft (April to early May)

- ☐ Attend the **interim reflection session**. Bring your results, graphs and any problems.
- ☐ Write the full draft using the structure below.
- ☐ Check the word count (4,000 or fewer) and that the question is the same everywhere.
- ☐ Hand the full draft to your supervisor by: .....

RRS prompt (after the session): What is the most important thing you need to fix before the final essay?

Structure and what goes in each section

SECTION	INCLUDE	WORDS
Title page	Title, research question, subject (Biology), word count. **No names**	not counted
Contents	Headings with page numbers	not counted
Introduction	Your question, why it matters, how you will answer it	about 300
Background and literature review	The biology your discussion needs, and what other studies found	700 to 900
Method	Variables, materials, procedure, safety and ethics, in enough detail to repeat	500 to 700
Results and analysis	Processed data, graphs, statistics, each explained	600 to 800
Discussion and evaluation	What the findings mean, compared with published work; how strong the evidence is and what limits it	800 to 1,000
Conclusion	A clear answer to the question, its limits, and next steps	about 400
References	Every source, in one consistent style	not counted

Writing that shows analysis and evaluation

TO DO THIS	TRY PHRASES LIKE
Point to evidence	“Figure 3 shows…”, “the mean rose from … to …”, “as reported by …”
Explain	“This is likely because…”, “this suggests that…”, “which would explain why…”
Compare	“This agrees with…”, “unlike the findings of…”, “whereas…”
Weigh the evidence	“The overlapping error bars mean…”, “with only five repeats…”, “the p value of … indicates…”
Judge a limitation	“This had a large effect because…”, “this is unlikely to change the conclusion because…”

Stage 8: Feedback and final essay (May to end of June)

Your supervisor gives written comments on your full draft. This is the only draft they can comment on in writing, and they cannot edit it for you.

Final checklist

- ☐ The research question is identical on the title page, in the introduction and in the conclusion.
- ☐ The title page has no student, supervisor or school name. Acknowledgements have no names either.
- ☐ Contents page with correct page numbers; page numbers on every page after the contents.
- ☐ Font size 12, 1.5 line spacing.
- ☐ Every figure and table is numbered, titled, referred to in the text, and credited if not your own.
- ☐ Organisms are written as *Genus species* in italics.
- ☐ Units and uncertainties are given for all measurements.
- ☐ Nothing important is hidden in an appendix or footnote.
- ☐ Every source is cited in the text and listed in the references, in one style.
- ☐ Any AI use is declared, following school rules.
- ☐ Word count is 4,000 or fewer.
- ☐ The file is under 10 MB.
- ☐ Final essay handed in by: ..... (before the summer holidays)

Stage 9: Viva voce and reflective statement (November, Year 2)

The viva voce is a short, friendly conversation with your supervisor about the finished essay. It celebrates your work and checks that it is yours.

Prepare by rereading your RRS and thinking about:

- ☐ the highlight of the research
- ☐ the hardest moment and how you dealt with it
- ☐ what you would do differently
- ☐ skills you have used since, in other subjects, CAS or plans for university

Then write your reflective statement (500 words or fewer) on the RPF. It is the only evidence for criterion E, and a blank form scores zero.

PARAGRAPH	WRITE ABOUT
1	How your question and approach developed, and the decisions behind the changes
2	A specific problem and what you learned from solving it
3	What you would do differently, and why
4	How you have grown as a learner, and where you now use these skills

The five criteria

CRITERION	MARKS	IN SHORT
A. Framework for the essay	6	A focused question, suitable methods, a clear structure
B. Knowledge and understanding	6	Accurate biology from good sources, in correct terms
C. Analysis and line of argument	6	Evidence-based analysis and a clear thread from question to conclusion
D. Discussion and evaluation	8	What the findings mean and how strong they are, weighed throughout
E. Reflection	4	How the EE changed you as a learner, with examples

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